

PINEGA, Russian Federation

WMO: 225630

Lat: 64.700N

Lon: 43.383E

Elev: 100

StdP: 14.64

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-23.2	-35.2	0.7	-30.2	-28.1	1.1	-23.1	15.8	19.6	14.1	18.6	2.0	330	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	81.2	65.6	77.2	63.8	73.4	61.9	68.1	77.5	65.9	74.1	63.7	71.2	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	92.9	72.2	62.7	85.5	69.7	60.3	78.6	67.3	32.5	77.5	30.7	74.3	29.1	71.0	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	13.1	11.7	-33.2	85.7	6.4	3.7	-37.8	88.3	-41.5	90.5	-45.0	92.6	-49.7	95.3
			-33.4	70.9	6.3	3.4	-37.9	73.3	-41.6	75.3	-45.2	77.2	-49.8	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	7.8	9.5	19.6	32.7	44.0	53.6	59.8	55.5	46.5	35.2	23.0	14.9
	DBStd	20.72	15.88	15.39	11.01	8.59	9.49	8.48	7.83	7.23	6.40	7.29	11.87	14.94
	HDD50	6735	1309	1133	943	520	240	54	11	25	141	460	810	1089
	HDD65	11511	1774	1553	1408	968	654	355	200	307	554	924	1260	1554
	CDD50	765	0	0	0	2	53	163	314	195	37	1	0	0
	CDD65	65	0	0	0	0	1	14	38	12	0	0	0	0
	CDH74	709	0	0	0	0	25	175	386	122	1	0	0	0
	CDH80	137	0	0	0	0	2	40	72	23	0	0	0	0
	WSAvg	5.6	5.4	5.5	5.8	5.9	6.3	5.6	5.3	5.1	5.2	5.9	5.7	5.7
	PrecAvg	21.9	1.2	1.0	1.1	1.3	1.5	2.7	2.5	2.8	2.3	2.2	1.9	1.5
(15) Precipitation	PrecMax	29.3	2.8	2.4	3.0	2.8	3.0	6.9	4.9	5.8	3.9	3.9	4.2	2.6
	PrecMin	16.7	0.3	0.1	0.2	0.3	0.2	0.0	0.9	0.6	0.4	0.3	0.7	0.5
	PrecStd	3.3	0.5	0.5	0.6	0.7	0.7	1.5	1.1	1.2	0.9	0.9	0.8	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.4	45.0	64.7	77.3	84.5	85.9	83.1	69.5	54.6	43.5	37.1
		MCWB	33.0	34.9	37.6	50.2	60.0	66.4	68.3	67.9	60.0	50.6	42.2	35.9
	2%	DB	32.6	33.8	38.9	54.0	71.0	79.2	81.9	77.2	64.0	50.1	40.4	34.8
		MCWB	31.8	32.8	33.9	44.3	55.7	63.5	66.1	65.3	57.1	47.5	39.6	33.6
	5%	DB	30.5	31.9	36.3	48.4	66.1	74.1	78.6	72.0	60.3	47.2	37.6	33.5
		MCWB	29.6	31.0	33.0	40.7	53.4	60.0	65.4	62.6	55.0	45.3	36.9	32.6
	10%	DB	27.7	29.2	34.2	44.7	61.3	69.6	74.9	68.0	56.9	44.6	35.1	32.1
		MCWB	26.8	28.1	31.8	38.5	50.9	58.2	63.8	61.3	53.4	43.2	34.1	31.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	35.0	38.0	51.7	62.5	68.3	72.6	70.4	61.3	51.7	42.8	36.0
		MCDB	33.7	35.9	43.3	62.1	74.5	81.2	81.6	78.8	66.8	53.4	43.3	36.5
	2%	WB	31.9	33.0	35.3	45.3	58.2	65.4	69.1	66.9	58.3	48.1	39.6	33.9
		MCDB	32.6	33.7	37.7	52.8	67.2	75.7	77.6	74.7	62.7	49.8	40.1	34.4
	5%	WB	29.6	31.0	33.5	41.6	54.7	62.6	67.0	64.1	55.9	45.5	36.9	32.8
		MCDB	30.4	31.9	35.5	46.8	64.2	70.8	75.4	70.2	59.4	47.0	37.5	33.5
	10%	WB	26.8	28.3	32.1	39.1	51.6	59.4	64.8	61.9	53.5	43.3	34.4	31.3
		MCDB	27.7	29.3	34.2	43.9	60.4	67.0	72.5	67.1	56.6	44.5	35.0	32.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	13.4	16.8	16.0	17.5	18.1	17.8	15.4	12.8	7.6	7.7	9.8
		MCDBR	9.9	9.5	14.6	21.7	27.1	25.3	24.4	22.0	19.5	10.1	6.8	6.6
		MCWBR	9.7	9.1	10.9	13.5	14.4	11.7	11.5	11.9	12.5	8.1	6.9	6.7
	5% WB	MCDBR	9.7	9.6	11.0	19.5	24.3	21.7	21.1	19.5	16.8	9.5	6.8	6.3
		MCWBR	9.6	9.4	8.7	13.1	13.8	11.4	11.6	11.9	11.4	8.2	6.9	6.5
(40) Clear-Sky Solar Irradiance	taub		0.220	0.281	0.310	0.345	0.350	0.347	0.386	0.360	0.331	0.305	0.356	0.217
	taud		1.935	2.061	2.112	2.157	2.422	2.484	2.383	2.435	2.503	2.347	2.576	1.947
	Ebn at Noon		92	187	239	259	272	276	259	255	240	187	89	26
	Edh at Noon		8	20	30	36	31	30	32	28	21	16	7	3
(44) All-Sky Solar Radiation	RadAvg		41	209	633	1171	1496	1664	1597	1089	595	213	52	14
	RadStd		5	38	81	139	130	155	167	115	60	36	10	1

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page